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RESEARCH PROTOCOL DOCUMENT

A Comparative Analysis of Readiness-Based and Completion-Based I/O Models in the Linux Operating System:
A Study of epoll and io_uring

Scoping Review Protocol — SALSA Framework
Search · Appraisal · Synthesis · Analysis

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Document Type	Scoping Review Protocol
Framework	SALSA (Search, Appraisal, Synthesis, Analysis)
Date	May 2026
Status	Under Review / In Preparation

This protocol is submitted in partial fulfilment of the requirements for Advanced System Research at the Comnet Laboratory, OAU. All findings are provisional pending full literature review.

TABLE OF CONTENTS

1.	Background and Context	3
2.	Rationale for Scoping Review	4
3.	Search Strategy	5
3.1	Central Research Question	5
3.2	Inclusion Criteria	5
3.3	Exclusion Criteria	6
3.4	Primary Search Databases	7
3.5	Search String	7
4.	Appraisal Framework	8
5.	Synthesis Plan	9
6.	Analysis Plan	9
7.	PRISMA-ScR Flow	10
8.	References	10

1. BACKGROUND AND CONTEXT

Modern Linux applications demanding high-throughput, low-latency I/O — including web servers, database engines, network proxies, and storage systems — have long relied on readiness-based I/O multiplexing, most prominently through the **epoll** interface introduced in **Linux 2.5.45 (2002)**. Under this model, the kernel notifies userspace that a file descriptor is ready for an operation; the application then issues the operation itself, incurring a syscall round-trip per I/O unit.

In 2019, Jens Axboe introduced **io_uring** (merged in **Linux 5.1**), a completion-based asynchronous I/O interface built around two shared-memory ring buffers: the **Submission Queue (SQ)** and the **Completion Queue (CQ)**. Under this model, applications submit I/O descriptors without blocking; the kernel processes them and posts completion events. Crucially, **io_uring** can execute entire I/O pipelines with **zero syscalls** via **SQPOLL mode** — a theoretically significant advantage at high concurrency.

The rapid evolution of **io_uring** — spanning basic file I/O in 5.1, networked I/O, fixed buffers, multishot operations, and kernel-side worker threads across versions 5.6 to 6.x — has produced a moving target for comparative analysis. This scoping review applies the **SALSA framework** to systematically map, appraise, and synthesise the available evidence on performance, scalability, and implementation complexity across both models.

Kernel Version	io_uring Feature Introduced	Significance
Linux 5.1 (2019)	Core ring buffer, basic file I/O	Initial release
Linux 5.6 (2020)	Networked I/O, SQPOLL mode	Zero-syscall operation enabled
Linux 5.11 (2021)	Fixed buffers, registered files	Memory-pinning, fd registration
Linux 5.19 (2022)	Multishot operations	Single SQE, multiple CQEs
Linux 6.x (2023+)	Direct descriptors, kernel workers	Full kernel-side pipeline control

Table 1: io_uring kernel evolution milestones

2. RATIONALE FOR SCOPING REVIEW

A scoping review (as distinct from a systematic review with meta-analysis) is appropriate for this inquiry for the following reasons:

- **Heterogeneous literature design.** The evidence base spans microbenchmark papers, full systems evaluations, kernel internals analyses, and runtime implementation reports — formats that resist pooled quantitative synthesis.
- **Emerging evidence base.** `io_uring` is six years old and its capabilities have changed substantially across kernel versions, making a definitive systematic review premature.
- **Landscape mapping objective.** The primary goal is to identify what is known, what is contested, and what is genuinely absent — rather than to produce a single pooled effect estimate.
- **Grey literature relevance.** Scoping reviews permit inclusion of kernel mailing lists, engineering blog posts (e.g., Cloudflare, Axboe design documents, Red Hat Developer Guide) that are invaluable in systems research but excluded from conventional systematic reviews.

Note: This protocol is pre-registered conceptually against the PRISMA for Scoping Reviews (PRISMA-ScR) checklist. A PRISMA-ScR flow diagram is provided in Section 7.

3. SEARCH STRATEGY

3.1 Central Research Question

Note: How do readiness-based and completion-based I/O models compare in terms of performance, scalability, and implementation complexity in modern Linux systems?

3.2 Inclusion Criteria

- **Publication Date:** Studies published from **2019 to present**. `io_uring` was introduced in Linux kernel 5.1 (May 2019); prior work serves as background only.
- **Study Type:** Empirical benchmarking, comparative experimental studies, system design papers measuring I/O model performance, and survey/review papers synthesising `epoll` vs. `io_uring`.
- **Primary Subject:** Papers where `io_uring`, `epoll`, or the readiness/completion I/O paradigm is a central focus — not a passing mention.
- **Operating System Scope:** Studies conducted on Linux kernel environments (any distribution: Ubuntu, RHEL, Arch, etc.).
- **Performance Metrics:** Papers reporting at least one measurable outcome: throughput (IOPS, MB/s), latency (mean, p99, tail), CPU utilisation, syscall overhead, or scalability under load.
- **Journal Quality Threshold (Q1 Mandate):** A minimum of **75% of all included primary studies must originate from Q1-ranked journals or equivalent top-tier conference venues**, as classified by Scimago Journal Rankings (SJR) or the CORE Conference Ranking (A* and A). For computer systems research, the following venues are considered Q1-equivalent by default: USENIX OSDI, USENIX ATC, USENIX FAST, ACM SOSP, ACM EuroSys, ACM SIGOPS Operating Systems Review, IEEE Transactions on Computers, IEEE/ACM International Symposium on High-Performance Computer Architecture (HPCA), and PVLDB. The remaining 25% may draw from Q2-ranked sources or credible grey literature, provided the appraisal score threshold ($\geq 6/10$) is independently met.
- **Language:** English only.
- **Accessibility:** Full-text availability required; abstract-only records are excluded.
- **Publication Venue:** Peer-reviewed Q1 journals (SJR-verified), top-tier conference proceedings (CORE A* or A rated), and credible technical reports from kernel contributors. Venue ranking will be verified at the time of full-text screening using the most current SJR and CORE listings available.

3.3 Exclusion Criteria

- **Pre-`io_uring` Literature (before May 2019)** — except foundational `epoll` background references.
- **Non-Linux Platforms** — Windows IOCP, macOS `kqueue` (without Linux comparison context), embedded systems.
- **Superficial Mention** — papers where `epoll`/`io_uring` appears only as a runtime dependency, never evaluated.
- **No Empirical Component** — purely theoretical/architectural papers making no performance claims (usable as background only).
- **Non-Peer-Reviewed Sources** — blog posts, Medium articles, Stack Overflow threads, vendor marketing materials as primary sources.
- **Duplicate Records** — keep only the most complete version (journal version preferred over conference preprint).
- **Irrelevant I/O Models** — DPDK, SPDK, BSD `kqueue`-only papers without Linux interface discussion.

- **Insufficient Detail** — papers not describing experimental setup, workload, or kernel version with sufficient reproducibility.
- **Below Q1 Threshold (primary source quota):** Papers from Q3, Q4, or unranked journals are excluded as primary sources. Q2 papers are admissible only within the 25% non-Q1 allowance and must independently satisfy the appraisal score threshold. Venue ranking is assessed at the time of full-text screening; papers in journals that have been downgraded from Q1 since publication will be treated according to their ranking at time of publication.

3.4 Primary Search Databases

Database	Coverage Focus	Access Mode
VLDB / PVLDB	Database internals, storage engines, transaction processing	Open access
ACM Digital Library	OS/systems papers — primary venue for SIGOPS, SOSP	Institutional
IEEE Xplore	Networking, hardware-adjacent, embedded systems work	Institutional
USENIX	OSDI, ATC, FAST — gold standard for systems research	Free / Open
Google Scholar	Citation chaining; forward/backward reference tracking	Free
ScienceDirect	Elsevier peer-reviewed systems journals	Institutional
Scopus	Elite citation indexer; multi-publisher mapping	Institutional

Table 2: Primary search databases with coverage focus

3.5 Search String

The following Boolean search string will be adapted per database syntax:

```
("io_uring" OR "io_uring interface" OR "completion-based I/O" OR "libaio" OR "POSIX AIO")
AND
("epoll" OR "readiness-based I/O" OR "I/O multiplexing" OR "event-driven I/O"
OR "kqueue" OR "event loop")
AND
("performance" OR "throughput" OR "latency" OR "scalability" OR "syscall overhead"
OR "IOPS" OR "tail latency" OR "context switch" OR "zero-copy" OR "polling mode")
AND
("Linux" OR "kernel" OR "NVMe" OR "storage" OR "network" OR "file system")
AND
("benchmark" OR "comparison" OR "comparative" OR "evaluation")
```

Date range: 2019–2026. **Language:** English. Backward and forward citation chaining will be performed via Google Scholar on all included primary studies.

4. APPRAISAL FRAMEWORK

Each included study will be independently assessed against the five appraisal dimensions below using a **0–2 ordinal scale** (0 = absent, 1 = partially met, 2 = fully met). Studies scoring below **6/10** will be excluded from primary synthesis and flagged as background-only references.

ID	Criterion	Operationalisation	Score
A1	Workload Realism	Rejects purely synchronous uniform microbenchmarks. Studies must implement mixed multi-threaded execution, variable message payloads, and realistic connection lifecycle states (e.g., slowloris conditions, sudden bursts).	0–2
A2	Storage Baseline Media	Isolates block layer testing environments. Studies executing on legacy HDDs or basic SATA SSDs distort io_uring advantage. Protocol prioritises papers deploying NVMe arrays with kernel-level polling modes activated.	0–2
A3	Kernel Lifecycle Phase	Enforces kernel version context mapping. Features such as IORING_SETUP_SQPOLL, fixed file registration, and direct descriptors changed the internal locking layout. Evaluation must separate pre-v5.6 from modern v6.x capabilities.	0–2
A4	Parity and Tuning Ethics	Verifies that both code pathways were compiled with equivalent optimisation depth (e.g., edge-triggered EPOLLET vs. naïve level-triggered configurations, balanced interrupt mapping, identical memory-pinning bounds).	0–2
A5	Code Provenance	Mandates verifiable repository code availability to allow execution-path inspection for hidden architectural anomalies. Preprints without linked repositories are penalised.	0–2

Table 3: Appraisal rubric — 5 criteria, max score 10

Note: Grey literature (kernel mailing lists, Axbœ design documents, Cloudflare engineering posts) will be assessed on a separate three-point rubric: (1) author credibility and identifiability, (2) technical specificity, and (3) verifiability against kernel commit history.

4.1 Q1 Journal Compliance Tracking

To enforce the **75% Q1 mandate**, venue ranking will be recorded for every included study at the full-text screening stage. A running compliance log will be maintained throughout the review using the tracking schema below. If the Q1 proportion falls below 75% at any point, additional targeted database searches restricted to Q1 venues will be triggered before synthesis proceeds.

Source Category	Ranking Tool	Admissibility	Max % of Corpus
Q1 Journal (SJR)	Scimago Journal Rankings	✓ Primary source	No ceiling — minimum 75%

CORE A* / A Conference	CORE Conference Rankings	✓ Q1-equivalent	Counts toward 75% Q1 quota
Q2 Journal / CORE B	SJR / CORE	✓ Within 25% allowance	Max 25% of final corpus
Q3 / Q4 / Unranked Journal	SJR	✗ Excluded as primary source	0%
Grey Literature	Separate 3-point rubric	✓ Within 25% allowance only	Counted within 25% cap

Table 4: Q1 compliance tracking schema — venue admissibility and corpus quota by ranking tier

Note: Venue rankings will be verified using the Scimago Journal & Country Rank portal (scimagojr.com) and the CORE 2023 conference ranking list. For conferences not listed in CORE, the ERA 2010 ranking or independent citation impact metrics ($h5\text{-index} \geq 20$) will serve as a proxy for Q1-equivalence, subject to supervisor approval.

5. SYNTHESIS PLAN

Given the heterogeneity of the evidence base, synthesis will proceed through narrative and tabular approaches rather than statistical pooling:

- **Performance trajectory mapping:** Constructing chronological performance profiles of both interfaces across kernel versions to reveal capability convergence or divergence.
- **Workload pattern matrices:** Tabulating performance outcomes by workload type (network I/O, block I/O, mixed), storage medium, and concurrency level.
- **Ecosystem adoption analysis:** Mapping runtime and framework adoption (Tokio, liburing, Netty, etc.) as proxy evidence for real-world performance acceptance.
- **Conceptual synthesis:** Weaving the readiness-vs-completion paradigm debate into a coherent theoretical model of syscall overhead and kernel execution state transitions.

6. ANALYSIS PLAN

The analysis phase will interpret synthesised evidence through the following lenses:

- **Gap identification:** Systematically documenting workload types, kernel version ranges, and hardware configurations that are absent or underrepresented in the literature.
- **Contested findings:** Explicitly flagging where evidence is conflicting and proposing methodological explanations (e.g., kernel tuning differences, benchmark design).
- **Hedged conclusions:** Drawing conclusions proportionate to evidence strength, distinguishing between well-supported findings and speculative extrapolations.
- **Research agenda:** Proposing targeted empirical studies to address the most critical identified gaps, particularly for African network conditions and sub-Saharan infrastructure relevant to the Comnet Laboratory research mandate.

7. PRISMA-SCR FLOW (EXPECTED)

Phase	Action	Expected N	Rationale / Criteria Applied
Identification	Database + grey literature search	~800	7 databases × adapted search strings
Deduplication	Remove duplicate records	~550	Cross-database deduplication
Title/Abstract Screen	Apply inclusion/exclusion by title & abstract	~150	Inclusion criteria 1–4; exclusion criteria 1–4
Full-Text Assessment	Full-text eligibility assessment	~60	All inclusion/exclusion criteria; appraisal rubric
Included	Final included studies for synthesis	~35–45	Appraisal score ≥ 6/10; full criteria met; ≥75% from Q1/CORE A*/A venues

Table 4: PRISMA-ScR flow — expected record counts at each screening stage

Note: Expected N values are estimates based on preliminary database scoping. Actual counts will be recorded and reported in the final scoping review manuscript per PRISMA-ScR guidelines.

8. KEY BACKGROUND REFERENCES

The following sources constitute foundational background and will not count as primary comparative sources. Full bibliography will be compiled during the synthesis phase.

- [1] Axboe, J. (2019). *Efficient I/O with io_uring*. Kernel Recipes / LKML Design Document.
- [2] Reimer, J. et al. (2023). Revisiting Network I/O APIs: The Readiness vs. Completion Debate. *Proceedings of USENIX ATC*.
- [3] Cloudflare Engineering Blog (2022). *io_uring in Production: Lessons Learned*. engineering.cloudflare.com.
- [4] Red Hat Developer (2021). *An Introduction to the io_uring Asynchronous I/O Framework*.
- [5] Corbet, J. (2019). Ringing in a new asynchronous I/O API. *LWN.net*.
- [6] Liao, X. et al. (2022). High-Performance Networking with io_uring. *IEEE/ACM International Symposium on Quality of Service*.
- [7] Linux Kernel Documentation (2023). *io_uring — Notes on io_uring*. kernel.org/doc/html/latest/.